Microsoft vs. S&P 500

A 3-Year Risk-Return Analysis | Project Overview

Analyst: Gizealew Tefera | **Period analysed:** 30 Oct 2023 - 27 Oct 2026 (751 trading days) | **Tools:** Python, pandas, NumPy, Matplotlib, yfinance

1. Project Objectives

This project evaluates whether holding Microsoft (MSFT) over the past three years rewarded an investor for the extra risk of owning a single stock, compared with holding the S&P 500 index. It is a standard equity-research question: a single-stock position should earn a premium that justifies the risk that diversification would otherwise remove.

Primary objective

Quantify and compare the return, volatility, risk-adjusted performance and downside risk of MSFT and the S&P 500 over the same period.

Supporting objectives

- Build a clean, validated daily price dataset from a public data source.
- Calculate standard performance metrics: cumulative return, CAGR, annualized volatility, Sharpe ratio and maximum drawdown.
- Visualize cumulative growth and drawdown history to show how each asset behaved through market stress.
- Translate the numbers into a clear conclusion, with the limitations of the analysis stated openly.
- Package the work professionally so it can be reproduced and reviewed by others (GitHub, portfolio, LinkedIn).

Key question

Did MSFT deliver enough additional return to compensate for its additional risk versus the S&P 500?

2. Headline Results

Metric	Microsoft (MSFT)	S&P 500
Cumulative return	61.21%	85.34%
Annualized return (CAGR)	17.32%	22.92%
Annualized volatility	26.15%	14.93%
Sharpe ratio (0% risk-free)	0.74	1.46
Maximum drawdown	-34.50%	-18.90%

Conclusion: over this window the S&P 500 delivered about 24 percentage points more cumulative return with roughly 43% lower volatility and a drawdown about half as deep. MSFT's Sharpe ratio was about half the indexes, so the single-stock position did not earn a premium for its extra risk.

3. Project Workflow

The analysis follows a five-stage workflow, from raw data to a documented conclusion.

Stage	What was done	Output
1. Data collection	Downloaded daily adjusted closing prices for MSFT and the S&P 500 index (^GSPC) from Yahoo Finance using the yfinance API, with adjustment for splits and dividends on MSFT.	751 daily observations per asset
2. Data validation and cleaning	Normalised timestamps and timezones, aligned both series on a common date index, and ran automated checks: data is non-empty, dates are unique, prices are strictly positive, and no values are missing.	Clean, validated dataset (0 missing, 0 duplicates)
3. Exploratory analysis	Computed descriptive statistics for MSFT price (range $329.59 to $543.45, mean $432.26, median $417.60) and plotted the price history.	Price summary and trend chart
4. Metric calculation	Calculated daily returns, cumulative returns, CAGR, annualized volatility (daily standard deviation x sqrt(252)), Sharpe ratio, and drawdown from a running-peak wealth index.	Performance summary table
5. Visualisation and reporting	Charted cumulative returns and historical drawdown for both assets, exported results to CSV, and documented findings, limitations and next steps.	Charts, README, this overview

Methodology notes

- **Cumulative return:** compounding of daily returns, (1 + r) cumulative product minus 1.
- **CAGR:** (1 + cumulative return) raised to 1/years, minus 1, using calendar years.
- **Sharpe ratio:** annualized mean return divided by annualized volatility, assuming a 0% risk-free rate.
- **Maximum drawdown:** the largest peak-to-trough fall in the wealth index over the period.

Limitations to keep in mind

- Three years represents one market regime; results should not be extrapolated to other horizons.
- The S&P 500 price index excludes dividends while adjusted MSFT prices include them, which slightly understates the index's total return.
- The Sharpe ratio assumes a 0% risk-free rate; a Treasury-bill series would be more precise.
- The analysis is descriptive. It shows what happened, not why.

4. Technical Skills Demonstrated

The table below maps each skill to where it appears in this project.

Skill area	Skills and tools	Evidence in this project
Financial analysis	Return and risk measurement, CAGR, Sharpe ratio, volatility, drawdown analysis, benchmark comparison, risk-adjusted performance	Full MSFT vs. S&P 500 comparison and investment conclusion
Programming	Python, Jupyter Notebook, reproducible scripts, dependency management (requirements.txt)	End-to-end analysis notebook that runs top to bottom
Data handling	pandas, NumPy, time-series alignment, timezone normalisation, API-based data collection (yfinance)	Merged and cleaned 751-day dataset for two assets
Data quality	Missing-value and duplicate checks, assertion-based validation, sanity checks on price data	Validation step before any calculations
Data visualisation	Matplotlib, Seaborn, time-series line charts, drawdown charts	Cumulative return and drawdown charts
Communication	Executive summary writing, translating metrics into conclusions, stating limitations, documentation	README case study, this overview, LinkedIn post
Project packaging	Structured repository, README, version control with GitHub, reproducibility instructions	Organised folder: notebooks, outputs, docs

5. Charts

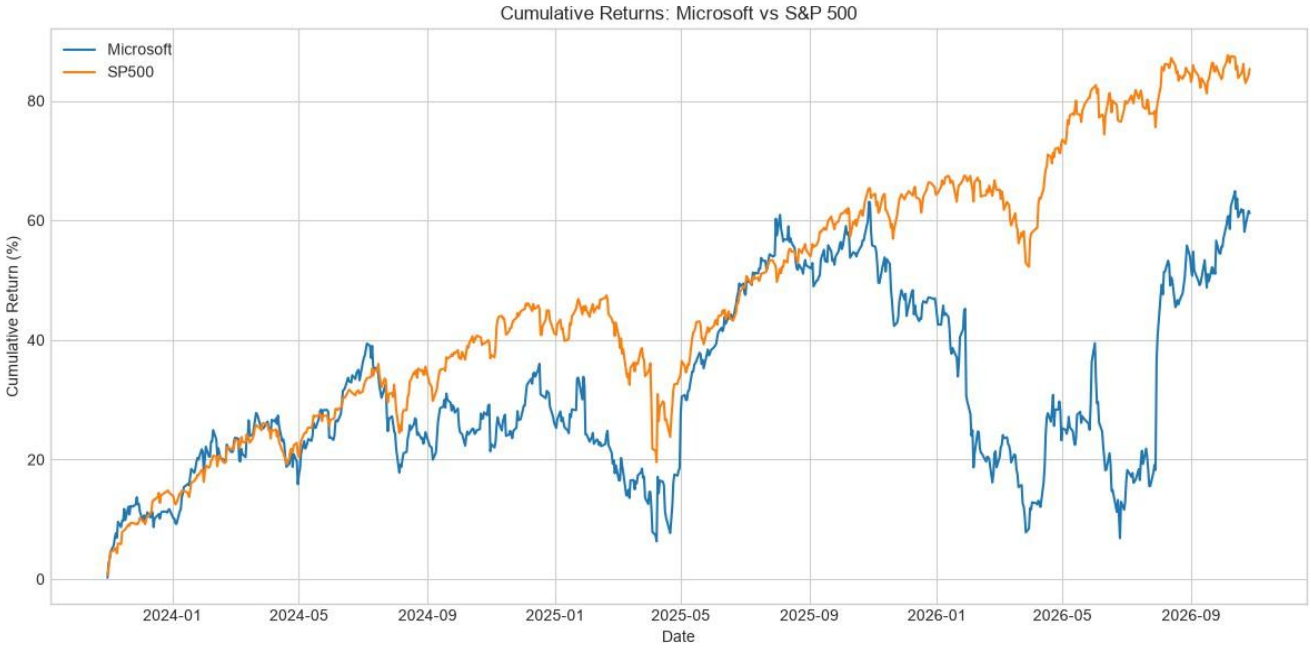


Figure 1. Cumulative returns: MSFT vs. S&P 500.



Figure 2. Historical drawdown: MSFT vs. S&P 500.

Disclaimer: this project is for educational and portfolio purposes only and is not investment advice.